

TEAM 359

WEBB.EXE

CHASING
INNOVATION



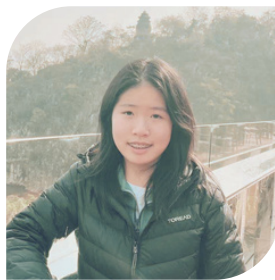
WE ARE COMMITTED TO
DEVELOPING INNOVATIVE, HIGH-
PERFORMANCE ROBOTS BY
PROMOTING EFFECTIVE
COLLABORATION AND CULTIVATING
DYNAMIC ENGINEERING
ENVIRONMENTS



JERRY HU

Captain, Mechanic
2nd year FIRST

Skills Developed: Team management, collaborative designing
Season Highlight: Designed novel and effective stringing method for reliable counter-stringing.



LINDA MENG

Captain, Programmer
2nd year FIRST

Skills Developed: Java Programming
Season Highlight: Created multi-threaded tele-op structure.



JAKE HUI

Mechanic, Builder
4th year FIRST

Skills Developed: Advanced CAD Design
Season Highlight: Designed new active intake, reduced cycle time by 25%



JONATHAN LI

Mechanic, Builder
4th year FIRST

Skills Developed: Fabrication, structural analysis
Season Highlights: Designed reliable end game climbing mechanism for consistent level 3 climb.



MAX LAN

Autonomous Programming
4th year FIRST

Skills Developed: Pedro Pathing Library, Sensor integration
Season Highlight: Incorporated new localization and pathing algorithm for successful auto



DAVID LIU

Builder, Human Player
3rd year FIRST

Skills Developed: Robot mechanism assembly
Season Highlight: Assisted to reinforce chassis structure to withstand high impact collision.



TINA HOU

Mechanic, Outreach
3rd year FIRST

Skills Developed: Community engagement, public speaking
Season Highlight: Connected with local non-profit and hosted programming courses for under-privileged communities



BOYANG LI

Outreach Lead
3rd year FIRST

Skills Developed: Fundraising, event coordination
Season Highlight: Secured a major sponsorship that provided critical funding



TYLER LIU

Game Strategy, Driving Coach
4th year FIRST

Skills Developed: Game analysis
Season Highlight: Designed optimal sample placement for bucket cycles



At Webb.exe, we value **diversity and inclusivity**. Our team is composed of members from **4 countries** across **3 continents**, and more importantly, of a **diverse level of FIRST experience**. Our team varies from **4th year** FTC participants, **VRC worlds finalists**, to just **2nd year** FTC participants. Our engineering processes, however, is not confined to a rigid hierarchy of sub-branches decided based on experience. We foster a collaborative development environment where everyone's voices are **equal and included** in all steps of designing, programming, outreach and more.

ABOUT US

We are an 18-year FTC team, **Team #359 "WEBB.EXE"** from The Webb Schools in Claremont, California. Our mission is to develop, inspire, and educate by fostering a culture of innovation and a passion for learning within our community and beyond throughout our participation in FTC. We strive to empower students from diverse backgrounds, promoting a prospering environment where every member can thrive and contribute.

Intelligent Control

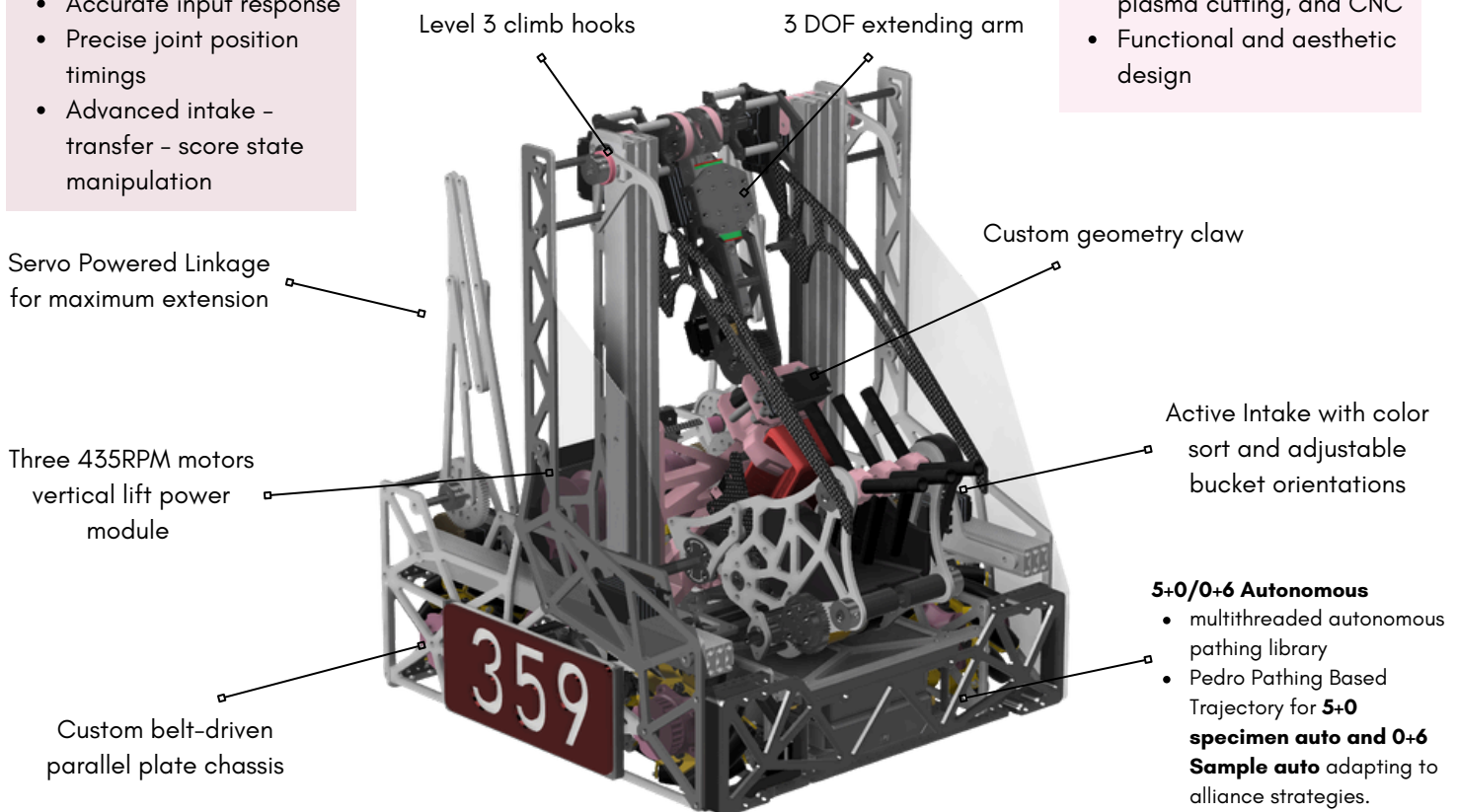
Multi-threaded teleop, which allows for:

- Accurate input response
- Precise joint position timings
- Advanced intake - transfer - score state manipulation

ROBOT OVERVIEW

Custom Parts

- 95% custom parts
- 3d printing, laser cutting, plasma cutting, and CNC
- Functional and aesthetic design



OUTREACH HIGHLIGHTS

- **Hosted** all SoCal League E meets at our school
- Mentored/Partnered with **40+ FTC** teams, **2 FRC** teams, **2 VRC teams**, and **1 FLL team**
- Met and collaborated with **Marine Biology Researcher, Mercedes Systems Engineer, and Harvey Mudd College Engineering and CS Professors, Professors and Senior Engineers at Shenzhen Polytechnic Institute** of Future Technology along with their college robotics team, **Director of AiRs Robotics Research Institute**
- Invited to attend the **Rockwell automation fair** as **FIRST ambassadors**

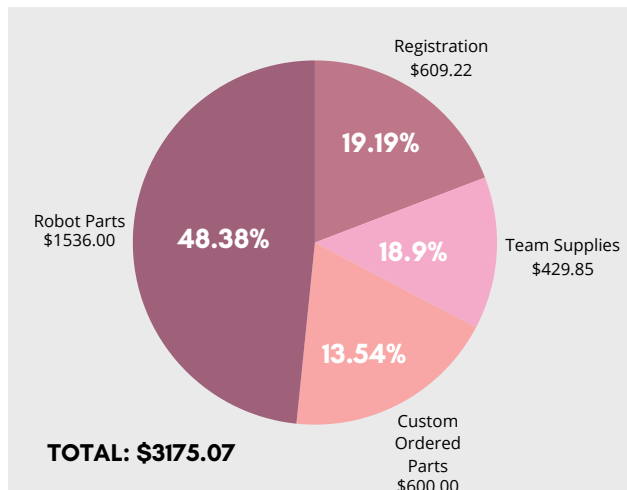
SEASON HIGHLIGHTS

- ILT 4 Winning Alliance
- ILT 4 Inspire Award 2nd Place
- 5 Specimen 50pts Autonomous
- 6 Sample 48pts Autonomous
- 200+ pts Solo Score
- College Acceptances

TEAM MANAGEMENT

As a small, independent team consisting of 9 members, we lack the resources to implement a complex, multi-faceted team management system. However, our team has developed a **unique organizational structure** that thrives on **our close-knit dynamic**. Built on the strong relationships formed through FIRST and other experiences, we naturally collaborate **to keep everyone engaged and informed about the season's progress** and individual responsibilities. This bond enables efficient communication, leverages each member's strengths, and allows us to adapt quickly to challenges **without relying on a rigid hierarchy**. By cultivating an environment where everyone is valued and involved, we not only make robotics an enjoyable experience but also optimize learning, creativity, and teamwork—**transforming each challenge into an opportunity for collective growth**.

BUDGET DISTRIBUTION



SPONSORS

We would like to express our heartfelt gratitude to our school and our generous sponsors for their unwavering support of our robotics team. Their contributions have provided us with the resources, opportunities, and encouragement needed to innovate, collaborate, and grow as inspiring FIRST participants.

WEBB

goBILDA
DREAM • MAKE • REPEAT

fw.



SHERATON



ROCKBROS



SendCutSend

PLANNING, EXECUTION, IMPORTANT DATES



Our team prioritizes outreach as a key means of building connections and giving back to communities in the FIRST way. We've reached out to teams and professionals, hosted events of varying scales and purposes, and worked to reach a diverse range of audiences. Our goal is to share our passion for engineering while making a positive, lasting impact on science and engineering communities.

MENTORS

- 400+ IN-PERSON OUTREACH HOURS
- 15,000+ INDIVIDUALS INFLUENCED
- 20+ EVENTS

This year, we've collaborated with several professionals to leverage their knowledge in support of our development. We are immensely grateful for their guidance over the season, which has been instrumental in helping us reach our achievements so far.

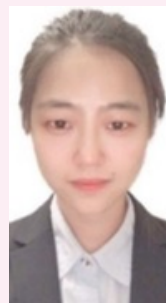
Dr. Bailey Jorgenson

- Marine Biology researcher from UC Santa Barbara
- Talked to us about marine biodiversity and its relationship to environment
- Mentored our submarine project on underwater mapping and fossil excavation



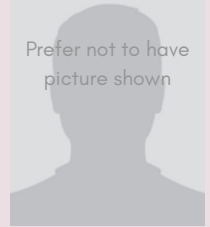
Samsung Electronics Yuexi Zhang

- PhD in CSEE, CV and Machine Learning researcher
- We explained the FIRST competition in a meeting
- Provided critical software tips in approaching the game



Tesla Senior System Validation Engineer Frank Liang

- Tesla battery developer
- Gave insights on mechanical structures and CV implementation logic



Abhishek Mantha

- Former Mercedes Benz autonomous driving developer
- Taught us essentials of autonomous driving and its real world implications



AI professor Ding Ning

- Chinese Professor of computer vision
- Provided experience and suggestions on pathing library and CV pipelines



Robotoics professor Duan Yi

- Chinese Professor of robotics and engineering from BUAA
- Conversed and exchanged ideas regarding design principles



Team Development & Sustainability

As the veteran team in our school's robotics program, we place great value on team growth and development throughout the season. We take pride in managing recruitment and training for our school's robotics program, dedicating significant efforts each year to bring new students into the FIRST community and guide them along the way. We also serve as the technical backbone of our school's introductory engineering and computer science courses. Through our commitment to mentorship, empowerment, and fostering a collaborative environment, we aim to inspire the next generation of innovators and problem-solvers within our school and the broader robotics community.



3/22/2024 - **Local** Outreach **300+** influenced

Admitted Students Day

- Hosted a station at event
- Showcased our CenterStage robot to prospective families

March 2024 - **Local** Outreach **30+** influenced

Submarine Project

- Built Submarine with Marine Biologist
- Collaborated with the Webb School's unbounded learning program

April 2024 - **National** Outreach **15** influenced

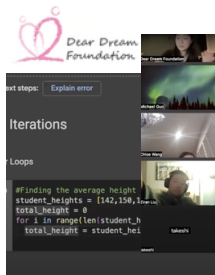
Team Meeting 15534 VERTEX

- Shared designs of CenterStage robot
- Gave Worlds advice
- Sat up long term partnership

July 2024 - **Regional** Outreach **100+** Influenced

Intro to Python Boot Camp

- Collaborated with local Non-profit Dear Dream Foundation
- Planned and taught 2 sessions of intro to Python summer Camp
- Aimed to provide STEM education resources to local under-resourced communities



August 2024 - **Local** Outreach **30+** Influenced

Innovation Seminar

- Helped design and open a new class for the Webb School
- Members on team took the class as official technical support for classmates with less experience



9/13/2024 - **Local** Outreach **500+** Influenced

Charles Huang Innovation Center

- Helped started the first official makerspace in the Webb School
- In charge of hardware selection and maintenance

11/3/2024 - **Local** Outreach **300+** Influenced

Discover Webb Day

- Hosted the robotics station, math, and science station
- Showcased past seasons robots and season progress to admitted families

11/23/2024 - **Regional** Outreach **10,000+** influenced

Rockwell Automation Fair

- Invited to Rockwell Automation Fair as FIRST ambassadors
- Presented to a group of professionals in an 8 hour exhibition

1/24/2025 - **International** Outreach, **10** Influenced

Team Meeting 19013 G-Force

- Connected with Indian team G-Force
- Gave advices on mechanical structures and designs

2/2/2025 - **Local** Outreach, **6** Influenced

Team Meeting 7767 Blueprints

- Invited Blueprints to our lab
- Practiced together and helped tune autonomous pathing

2/12/2025 - **National** Outreach **12** Influenced

Team Meeting 15534 VERTEX

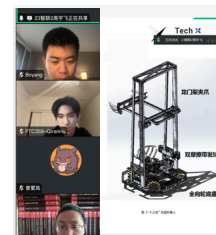
- Second meeting with VERTEX
- Shared Outreach experience, CV, and autonomous pathing library
- Discussed Future Outreach events

2/14/2025 - **International** Outreach **15+** Influenced

Team Meeting Chinese College

Robotics Challenge

- Learned about College robotics
- Gave some insights for the college team's robot design
- Sat up long-term collaboration for mutual mentorship



2/15/2025 - **International** Outreach **17** Influenced

Team Meeting 19600 Treeman

- Talked about outreach resources
- Discussed future outreach collaboration

2/15/2025 - **International** Outreach **17** Influenced

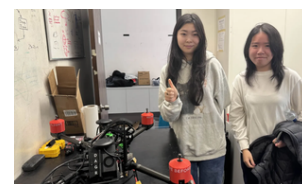
Team Meeting 13368 Mechatronics

- Gave suggestions on robot designs
- Established Long-terms partnership for international connections

2/20/2025 - **Regional** Outreach **8** Influenced

UCI Drone Club Visit

- Visited UC Irvine Drone Club ran by FIRST alumni
- Learned about college level engineering projects



2/27/2025 - **Regional** Outreach **9** influenced

Team Meeting 4348 RoboKnights

- Connected with 4348 RoboKnights online
- Talked about robot design and game strategy

3/3/2025 - **International** Outreach **250+** Influenced

Coast to Coast to Coast

- International online seminar co-hosted with 19600 and 15534
- Connected teams from west coast, east coast, and beyond the coast
- Shared environmental protection insights

SEASON GOALS

- Extend efforts to alumni
- Giving back to FIRST community
- Connecting with our league
- Skills Development
- Mentoring Teams
- Fostering STEM interest

Charles Huang Innovation Center September 13, 2024

Driven by a desire to improve the engineering environment at our school, members of our team played a key role in the development of the new Charles Huang Innovation Center. This is our school's first dedicated maker space, equipped with high-quality 3d printers, a laser cutter, a robotic arm, and more. The space serves as the foundation for the school's new project-based class, Innovation Seminar, and supports a variety of personal projects, helping to cultivate a strong STEM education.



Rockwell Automation Fair November 23, 2024 Submarine Project March, 2024

Our team had the incredible opportunity to represent FIRST at the 2024 Rockwell Automation Fair. During this event, we introduced FIRST Tech Challenge (FTC) and this year's game to a diverse audience of over 10,000 engineers and industry professionals in an 8 hour exhibition.



In the fall of 2024, we initiated a collaboration with the Raymond M. Alf Museum of Paleontology on an underwater fossil excavations project in Turlock Lake. Through an extensive R&D process we developed a manually controlled submarine capable of underwater mapping and fossil detection. The project was then integrated into our school's unique Unbounded Learning program. We brought a group of approximately 30 interested students on a week-long trip, where they designed, built, calibrated, and operated the drone submarine to collect and analyze measurements, fostering a deep appreciation for science and the environment.



Community Connections

Our team is committed to expanding the robotics program within our community. To further develop our engineering and programming skills, we visited the nearby Claremont Colleges. We also connected with the Webb Admissions Department, engaging with 5 alumni and 20+ prospective students. During this time, we shared our past season's engineering portfolio and game footage to promote FTC to those interested in joining the community. Additionally, we hosted robotics, math, and science stations during open house events, where we had the opportunity to speak with over 100 prospective families about the aforementioned programs.



League Connections

We have a longstanding tradition of hosting the Southern California League E. This year, we hosted Meets 0, 1, 2, and 3 at the Webb Schools. In addition to handling setup, cleanup, parking management, refereeing, and serving as MCs, we also actively engaged with our local community. We participated in scrimmages and invited local teams to visit our lab, offering them the opportunity to practice and learn from our engineering processes.



MENTORING

As a team formed mainly of upperclassmen, we've all been through and value the development of rookie teams. In our school, we recruited members of all rookie teams and mentored them along the way. Impacting the community, we committed to mentoring and starting teams in FTC and VRC. Globally, we are active in social media and online FIRST communities and give technical support to teams in China, Canada, Bangladesh, and more.

FTC 358 GaulBots & FTC 19888 Alfa-Tech

- School sibling team we've recruited and mentored along the season
- Taught Cad, Prototyping practices and Programming



FTC 11347 CircuitRunners Robotics

- Gave feedback to CAD
- Provided examples and inspiration for intake system

FTC 24748 Team Atom

- Taught CAD to team in Kazakhstan
- Introduced team to Fusion 360



FTC 19013 G-Force

- Exchanged approach in advanced CAD
- Gave insights to sub-system modification

FTC 23929 Grizzly Tech

- Donated parts
- Mentored on autonomous pathing, helped tune road runner
- Introduced team to Pedro Pathing

FTC 13368 Mechatronics

- Suggested professional engineering processes
- Supported team in CAD and Programming
- Shared Competition experiences

FLL 60444 Slipper Guys

- Helped found the team
- Mentored in person in Canada
- Taught advanced programming



VRC 72711P Σ^3

- Helped developed PID library for autonomous
- Tagged along their Signature event in UC Berkley

VRC 7700V Rolling Robotics

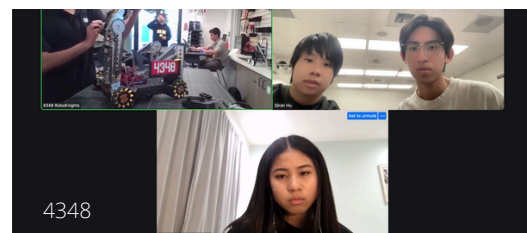
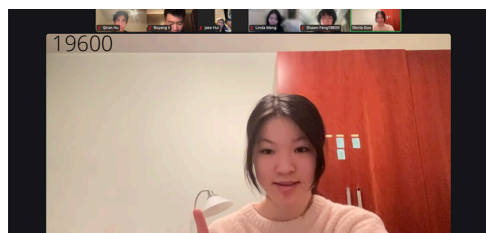
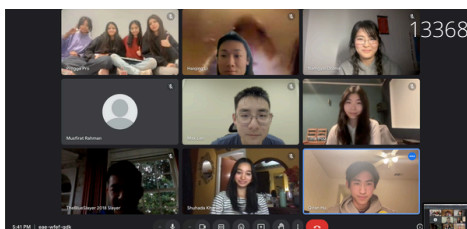
- Helped with interview and designing
- Aided in Autonomous driving Tuning

CONNECTED AND CONVERSED

- FTC 15534 VERTEX
- FTC 19600 Treeman
- FTC 7767 Blueprints
- FTC 26506 Axle-lotls
- FTC 9614 Hyperion
- FTC 4348 RoboKnights
- FTC 11770 Curiosity
- FTC 21380 Beyond Robotics

CONNECTING

Aiming to form strong relationships and make presence in the FIRST community, we invited many teams to online and in person team meetings in order to exchange season experiences. Through these meetings we've built robust connections and co-hosted bigger outreach events with these teams.



UPCOMING EVENTS

Our season does not end when competition ends. Despite our team all committed for States Championship preparation, we have already started to plan many upcoming outreach events for the off seasons.

- Official International Parts Sharing Platform for Environmental protection
- Connecting with LA government to discuss environmental notions
- Bike rides for marine garbage awareness
- Hackathons in China and Bangladesh
- Online Seminars for local Community with industry professionals
- Engineering and Coding courses with local Non-Profit
- Women in STEM events with Harvey Mudd College CS Professor
- Collaboration with education sector in Bangladesh

OUR DESIGN PROCESS

Having a planned design process is crucial for efficiently designing a high-performance robot. It allows us to systematically approach our design while considering all necessary factors and constraints. Under our unique team structure, each engineering decision is a collaborative result that considered the ideas of every member on the team.

1. Define the Problem

After developing a thorough understanding of the game, we define the task that the part or subsystem must accomplish.

2. Specify Requirements.

The objective of this step is to identify and define all constraints comprehensively. This ensures that potential issues during the implementation phase are minimized or avoided altogether.

- Are there any size or weight limitations for this part/subsystem?
- If applicable, how many motors or servos can it use?

3. Brainstorm ideas

Look back at previous seasons for potential inspirations on similar tasks such as intakes and lifts. Research real-world problems that are similar and their solutions. Conducting research on real-world problems and their solutions, such as car suspension systems or crane arms, can provide valuable insights for optimizing the robot's design and functionality in the current robotics competition.

4. Develop preliminary design and quickly fabricate

Nothing fancy is needed in CAD besides a part that can be quickly manufactured using processes like laser cutting and 3d printing or built from standard parts. The prototype's purpose is not to be final but to enable us to test and determine its viability before committing further.

5. Test prototype

We test for:

- Effectiveness: does it solve the problem it was designed to?
- Weaknesses: what are the problems with the design?
- Practicality & Feasibility: similar to effectiveness. Is the design overcomplicated? Hard to manufacture?

6. Analyze Results

After testing, we reevaluate the design (how well can we optimize it?). Is it worth pursuing, or should we try a different approach?

7. Repeat from step 4 until we are ready to move on

We continue the prototyping and testing cycle until we feel that we have a design that we are confident in. We should have already improved it significantly from the initial prototype and should have learned enough from testing and analysis to make a final implementation.

8. Final Implementation in CAD

After improving the initial prototype and learning from testing and analysis, we can proceed with the final implementation. See the CAD organization page for how we keep track of our whole assembly.

SEPTEMBER, 2024

BREAKING DOWN THE GAME

We watched the kickoff video and read Game Manual Part 2 to understand the game and scoring tasks. Using cost-benefit analysis, we prioritized tasks and planned subsystem development.

Key Functional Requirements (in order of R&D importance)

Robot is able to score both specimens and samples, extend into the submersible and be able to pick out the right color samples. Additionally is able to easily intake game elements from the human player station, both on the floor and grabbing from the wall.

Subsystems required to meet Key Functional Requirements

Drivetrain, lift system, intake, extension mechanism

OCTOBER, 2024

BRAINSTORMING

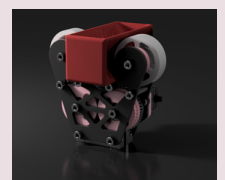
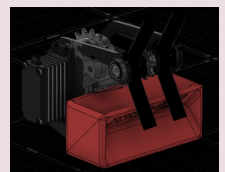
After understanding the game and design needs, we brainstormed, drawing from past robots, choosing a horizontal extension crane design with a passthrough mechanism. We also explored intake options like a compliant wheel and traditional claws.



NOVEMBER-DECEMBER, 2024

PROTOTYPING

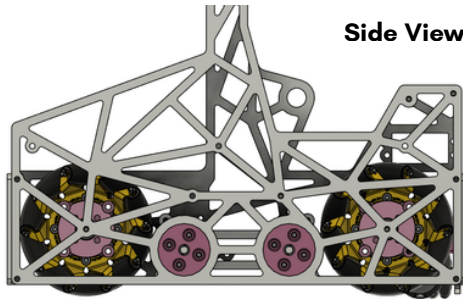
After brainstorming, we created CAD models of potential designs for testing in our next work session. Our focus was prototyping intake mechanisms, as their performance would heavily impact other systems. Through testing, we selected a traditional claw design. While slower than an active intake, it proved more reliable, robust, and easier to integrate into our autonomous path, meeting our functional and design goals.



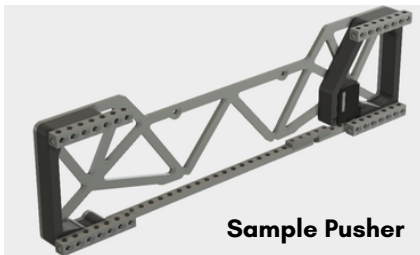
DRIVE TRAIN OVERVIEW

- The chassis features a custom parallel plate construction with a **belt-driven drivetrain** for **smooth and efficient motion**
- Equipped with 435 RPM motors and a 1:1 pulley ratio, the drivetrain achieves a **maximum speed of 0.696 m/s**
- The belt-drive system ensures smooth power transmission while minimizing maintenance requirements
- This chassis serves as the **foundational structure for key robotic modules**, including the intake, arm, horizontal extension, and lift
- To **mitigate backlash**, adjustable **belt tensioners** have been incorporated into the design
- A **sample pusher** is integrated at the chassis base to enhance responsiveness to suboptimal randomization conditions

Side View

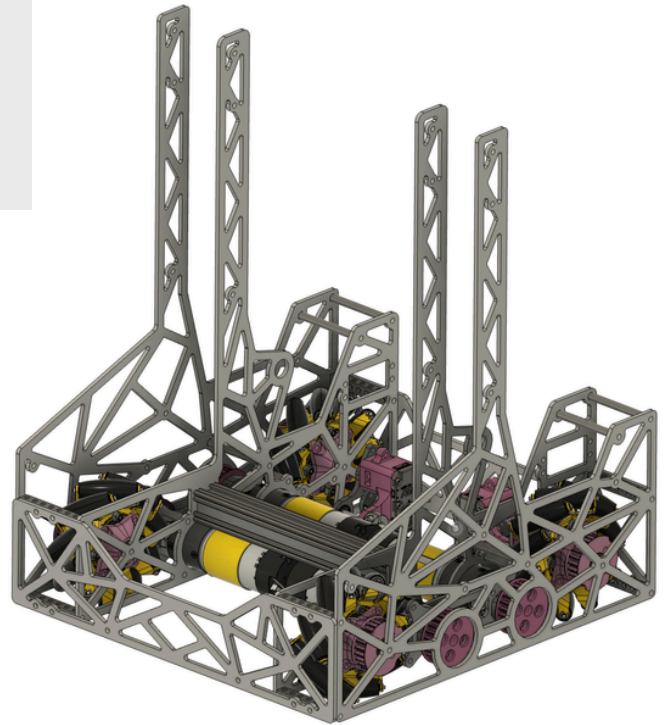


Sample Pusher



FUNCTIONAL REQUIREMENTS

- Integration of all subsystem mountings into 4 parallel plates, reducing excess weight.
- Consistent and adjustable belt tensioning to prevent backlash.
- Lightweight and aesthetically pleasing design for optimal performance and visual appeal.



V1 DESIGN

October 15, 2024



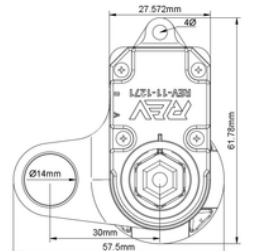
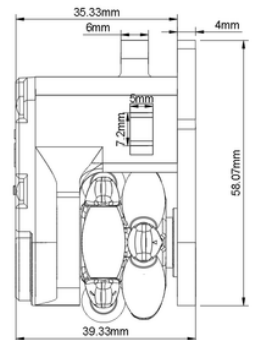
V1 Chassis



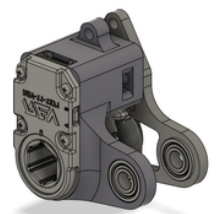
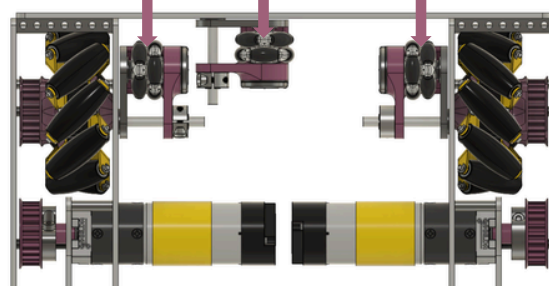
The previous belt driven system

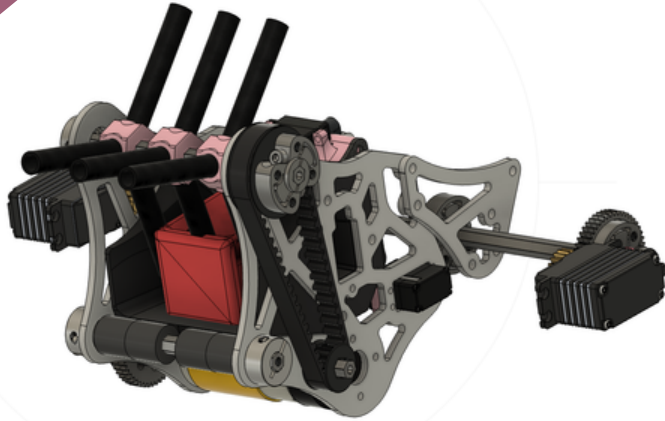
ODOMETRY

Leveraging open-source innovation, our team has engineered cost-effective odometry pods "TinyOdo" (Figure Right), priced competitively at \$65, which provide double the Clicks Per Rotation (CPR) compared to GoBilda's \$100 alternatives. Adhering to design requirements, our pods boast a smaller turning radius, surpassing other models in agility and performance. Central to our navigation system is the implementation of 3-wheel odometry, offering enhanced accuracy and control (Figure Bottom). This aligns seamlessly with our robot's overarching architecture.



Odometry Pods





INTAKE OVERVIEW

- **Compact drop-down** active intake designed to intake **all possible sample orientations** from the submersible
- Color sensor is integrated for **automatic sample color sorting**
- **Belt-driven** power transmission powered by a bare 6k rpm motor, providing reliable power to the intake system.
- Surgical Tubing runs on a **2k rpm bottom roller at 1k rpm**, driven by a **bare motor with a 2:1 ratio** for efficient movement.
- **Tailored components** such as custom bucket and aluminum plates designed to optimize the intake and handling process.

FUNCTIONAL REQUIREMENTS

- Quick and efficient intake of specimens from the ground.
- Capability to intake specimens from the wall.
- Ability to intake specimens from any orientation for flexibility.
- Lightweight and easily maintainable design for long-term reliability.
- Durability to withstand heavy use and frequent operation.
- Minimized time spent aligning, and enhancing operational efficiency.
- Reduced potential for human error, ensuring smooth and reliable performance.



ALIGNMENT LOCK

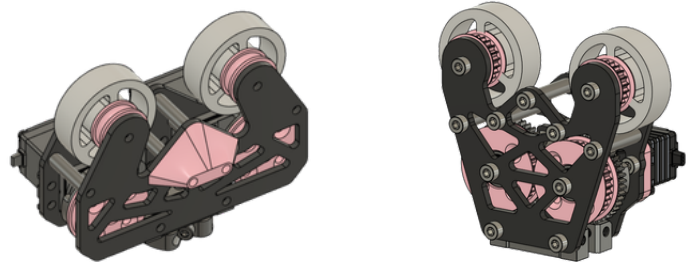
Facing inconsistencies with our previous iterations of an active intakes, we pivoted towards designing a mechanism that can ensure a reliable transfer. Our design utilizes a single servo, controlling the motion of both the lock and the entire bucket with a passively sprung arm. By rotating the bucket we can ensure that the sample is in the same position every single time for consistent transfers.

LEARNING PROCESS

Our intake process has proven highly efficient for collecting game elements from the field. Earlier in the season, we experimented with a claw pass-through mechanism to mitigate inconsistencies; however, we encountered challenges with alignment precision and cycle speed optimization. Consequently, we reverted to a dropdown active intake with a locking alignment system that reliably transfers elements to our end effector, ensuring faster cycle times overall.

V1 DESIGN

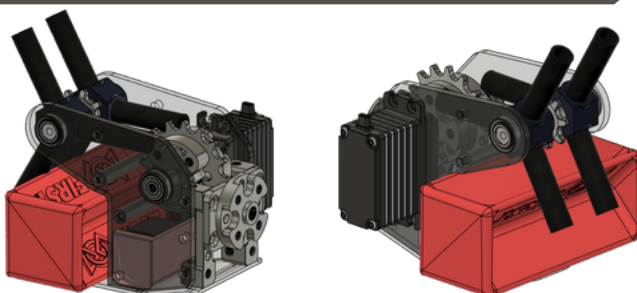
September 17, 2024



- 2 2.5-inch compliant wheels
- Two servos to single servo
- System is unreliable due to inconsistencies in servo power

V2 DESIGN

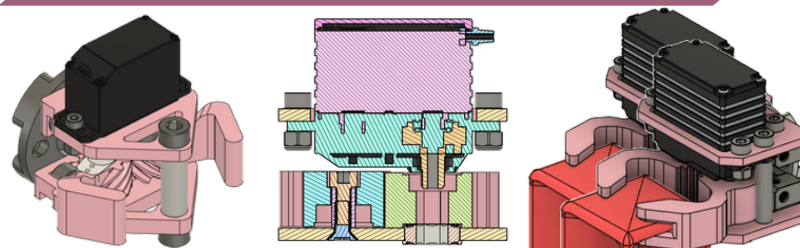
November 29, 2024



- Active intake design involving surgical tubing.
- Fastest intake speed.
- Geared-up servos and a bare 6000-rpm motor.
- Issues with integrating it into our passthrough system that have limited space.

V3 DESIGN

December 20, 2024



- Simpler, lightweight claw to be the most reliable.
- Single-servo geared arm claw for both micro 9kg and standard 25kg servos
- Alignment geometry from our initial prototyping.
- Allowed a 30° intake margin of error and effective sample collection from clusters.

EXTENSIONS OVERVIEW

Vertical Lift

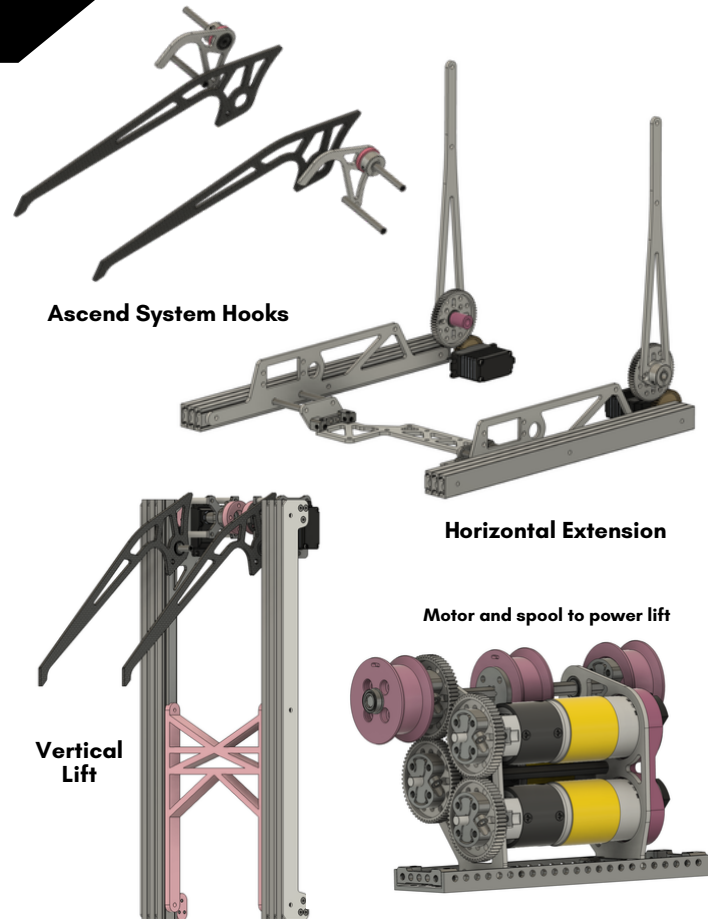
- Raises and lowers the robot, enabling the lift assembly to score **high bar specimens and high bucket samples**.
- Powered by three 435rpm GoBuilda motors on a 44mm spool, achieving a **maximum lift speed of 0.306 m/s**.

Horizontal Extension

- Reduces drive distance during cycles, allowing the intake to **reach more than two-thirds of the submersible zone**.
- Utilizes three stages of **Misumi Sar230 linear slides**, powered by two **Axon MAX servos** connected to linkages.
- Provides precise control of the extension position, enhancing operational accuracy and flexibility.

Ascend

- Our climb system shares the same power module as our lift, using three 435 rpm motors to support **up to 35 pounds of weight**. During the climb, the first set of active hooks pulls the robot upwards and locks onto a pair of sprung passive hooks. The **weight distribution is intentionally designed** to tilt the robot slightly backward, allowing the active hook to secure onto level 3, then repeat the motion to complete a successful level 3 climb.



LEARNING PROCESS

While stringing the lift slides, we discovered that a counter string—one that actively pulls the lift downward—is critical for stable high-speed operation. Accordingly, our lift power module incorporates three spools on a separate axis from the three motors. One spool, wrapped clockwise, exerts force to raise the lift, while the middle spool, wrapped counterclockwise under tension, provides the necessary pull downward. This configuration ensures proper tension, minimizes slack and maintains precise control at higher lift speeds.

FUNCTIONAL REQUIREMENTS

Horizontal Extension

- Allows for precise control of extension position.
- Enables the extension to reach deep into the submersible.

Vertical Extension

- Strong and stable to support the entire robot.
- Capable of reaching the top bucket.
- Operates at max possible speed to optimize cycle time.
- Integrates with level 3 ascend system for enhanced functionality

V1 DESIGN

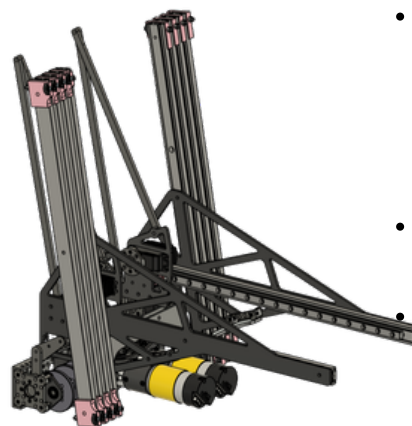
October 21, 2024



- The initial design was a geared double reverse 4-bar linkage powered by 2 servos mounted near the front of the robot
- It allowed for maximum possible extension range from a servo powered linkage
- The side plates serve as the attachment point for both the vertical slides and horizontal extension components
- The limitations of this design were:
 - the stability and reliability of the complex linkage system
 - the excess strain put on the servos from the large mechanical disadvantage

V2 DESIGN

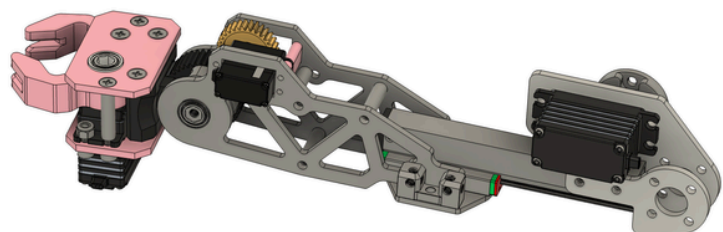
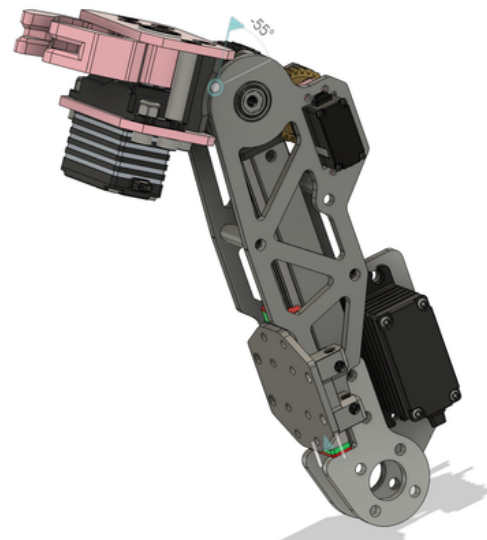
November 22, 2024



- The extension servo mounts were moved to the lower back of the lift side plates, concentrating the center of gravity of the robot and allowing for a simpler linkage design
- The horizontal extension was changed to a simple two-stage linkage
- This new approach greatly reduces servo strain and structural complexity, with only a slight reduction in maximum reach
- Vertical extensions still counterstrung to eliminate risk of lift jams
- The lift used thick 2mm diameter kevlar string rather than thin fishing line for extra reliability due to lower risk of jams. Pass through system was consistent yet had limited cycle speed.
- Due to the weight of the subsystem, when extended had a tendency to tip

END AFFECTOR OVERVIEW

- Asymmetrical parallel plates designed for a **compact, rigid structure**, minimizing the arm's footprint while maintaining structural stability.
- Custom claw geometry engineered for secure gripping, precisely tailored to **accommodate samples of varying shapes and sizes**.
- 3D-printed geared wrist joint provides precise control, delivering **accurate angular positioning** for smoother manipulation of specimens.
- Integrated linear rail offers **extended reach**, enhancing the robot's ability to score specimens by providing additional extension.



FUNCTIONAL REQUIREMENTS

- Wrist joint for precise orientation when scoring samples.
- Integrates seamlessly from the center of the chassis to the intake assembly.
- Sufficient length for flip-up scoring of specimens without requiring the vertical extension.
- Reliable claw geometry for firm and secure transfer of specimens.

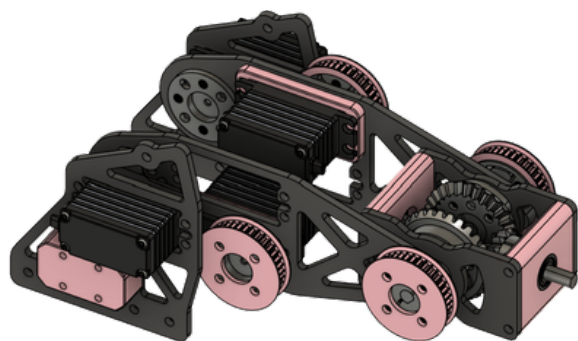
LEARNING PROCESS

Transitioning from a pass-through system to a transfer system, we were able to leverage our existing intake claw designs for the new end effector. The custom geometry proved both efficient and stable, allowing for reliable scoring of specimens and samples in precise orientations. Integrating 3D-printed components with custom plates resulted in a simple yet robust design, ensuring a secure fit and smooth operation.

V1 DESIGN

October 28, 2024

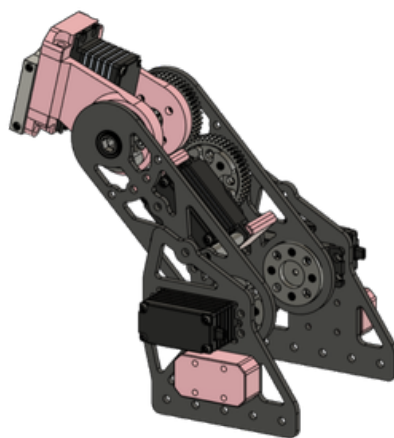
The original design of the arm of the robot had 3 degrees of freedom, but the pivot and wrist joints were powered using a belted bevel gear differential driven by two servos mounted in the arm. This design was chosen initially because it would allow for 2 servos worth of power in each joint due to the differential, but during prototyping, the added complexity caused major backlash issues, which prevented accurate control.



The two servos are mounted inside the arm, and are offset to both fit inside the tight space. The power is delivered to the differential by GT2 Timing belts on a 1:1 ratio.

V2 DESIGN

January 16, 2024



This iteration maintains consistency with the robot's overall parallel plate construction method. Cable management has significantly improved over V1. All cables are now shielded and supported by hard plastic at servo connection points.

The intake arm used for ILT features three specialized joints (base, pivot, and wrist) offering 3 degrees of freedom, powered by 4 Axon Max servos. Key improvements from V1 include:

- The base joint uses 2 Axon MAX servos mounted to the linear system with aluminum plates that also attach to the extension linkage.
- The pivot joint uses 1 Axon MAX servo with a GoBuildA 32mm ID Bearing for additional support on the servo horn, necessary for handling the forces when scoring specimens.
- The wrist joint uses 1 Axon MAX servo with a 1:1 gear ratio, positioned inside the arm for a more compact design.

OPMODE OVERVIEW

- Multithreaded code structure allows for separate **asynchronous loops** to handle automated sequences by using four separate control loops (drive, arm, extension, and input) running in parallel for smooth, simultaneous operation
- State-Based Logic which uses a **state machine** (e.g., "idle," "ground," "bucket," "wall") to ensure consistent, predictable movements across different robot modes when controlling complex subsystems
- Uses **non-linear exponent scaling** on joystick values, giving drivers precise control at low speeds and full power at higher stick deflections

```
TaskThread driveThread = new TaskThread(new TaskThread.Actions() {
    @Override
    public void loop() {--
    }
});

TaskThread armThread = new TaskThread(new TaskThread.Actions() {
    @Override
    public void loop() {--
    }
});

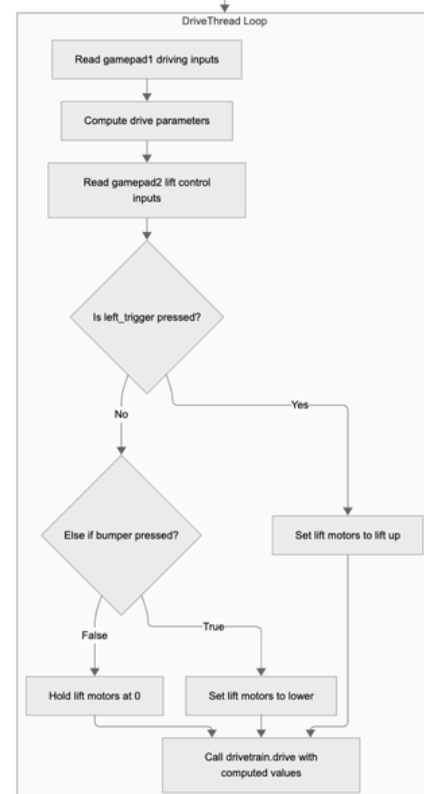
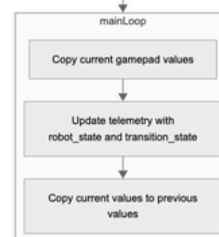
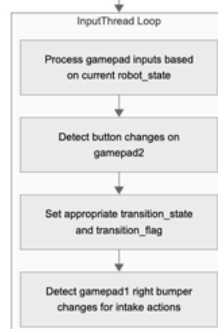
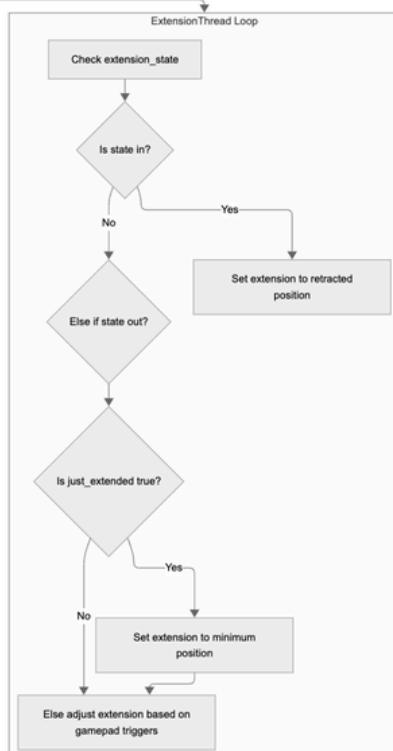
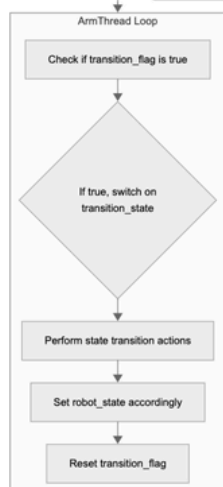
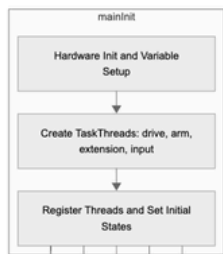
TaskThread extensionThread = new TaskThread(new TaskThread.Actions() {
    @Override
    public void loop() {--
    }
});

TaskThread inputThread = new TaskThread(new TaskThread.Actions() {
    @Override
    public void loop() {--
    }
});

registerThread(driveThread);
registerThread(armThread);
registerThread(extensionThread);
registerThread(inputThread);
```

FUNCTIONAL REQUIREMENTS

- Stable control of the drivetrain to allow for fast movement across the field and precise control when intaking and scoring
- Efficient State Transitions to allow drivers to move quickly between arm or intake positions with minimal button presses
- Automated sequences such as the transfer sequence to be executed with a single button press
- Safe Operation by ensuring servo positions and motor powers do not conflict



LEARNING PROCESS

We used a traditional single threaded, sequential structure for our league meets, but decided to switch over to multithreading for ILT. While multithreading made controlling many subsystems easier, the more complex code structure came with its own issues. We chose to handle all button inputs in a separate thread to properly detect double presses and prevent debounce, which resulted in issues with the different loops running out of sync. This issue was fixed by copying our current and past inputs and reading those instead of the actual button inputs on the controllers.

```
@Override
public void mainLoop() {
    // copy gamepad inputs
    currentGamepad1.copy(gamepad1);
    currentGamepad2.copy(gamepad2);

    // update previous gamepad states
    previousGamepad1.copy(currentGamepad1);
    previousGamepad2.copy(currentGamepad2);

    telemetry.addData("robot state: ", robot_state);
    telemetry.addData("transition state: ", transition_state);
    telemetry.update();
}
```

Autonomous Overview

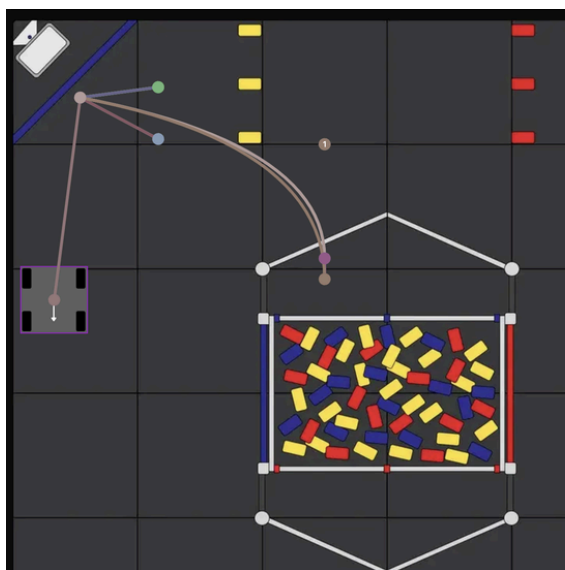
The autonomous code for our robot is designed to optimize path planning and execution using Pedro Pathing, a tool that leverages Bézier curves for smoother and more efficient movement. This allows for better error correction and ease of implementation compared to traditional splines. Additionally, we incorporated NextFTC, a command-based system, to streamline complex sequences by replacing switch statements with path-driven automation.

Functional Requirements

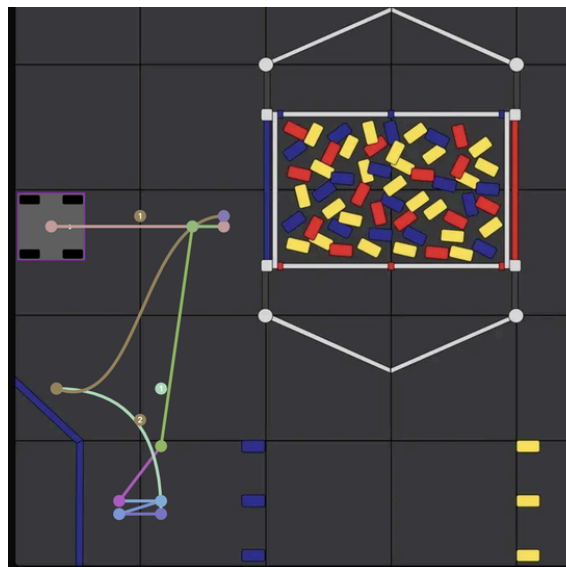
- Implement Pedro Pathing for autonomous navigation using Bézier curves.
- Integrate NextFTC for command-based execution of sequential robot actions.
- Ensure efficient handling of intake and transfer system delays without requiring a finite-state machine.
- Improve pathing accuracy and speed while maintaining ease of use.
- Follows paths with maximum motor power to complete 5 spec and 6 sample in time

While Road Runner has been the standard for autonomous pathing for the past few years, we decided to experiment with Pedro Pathing this season. Pedro Pathing uses Bezier curves which help increase speed while also making error correction smoother. We found Bezier curves, which are defined through control points rather than tangents, to be easier to work with than Road Runner's splines. In addition, Pedro Pathing offers an excellent browser-based visualizer that makes path planning fast and easy.

5 Sample Auto Path



5 Specimen Auto Path



Command Based Autonomous Routine

Our intake and transfer systems require a lot of time-based delays between actions, which cannot be implemented efficiently in a finite-state machine. As a result, we use a command-based system, which organizes all autonomous actions such as servo movements and path following into commands. These commands are called sequentially and in parallel using wrappers, which allows for complex simultaneous movements with multiple PIDF controllers and servo sequences.

```
public Command sampleAuto() {
    return new SequentialGroup(
        new ParallelGroup(
            new FollowPath(line1, true),
            transfer()
        ),
        new Delay(0.15),
        new ParallelGroup(
            new FollowPath(line2, true),
            Arm.INSTANCE.extendB1()
        ),
        new Delay(0.15),
        Intake.INSTANCE.down(),
        new Delay(0.1),
        Intake.INSTANCE.spinIn(),
        new Delay(0.2),
        new FollowPath(line3, true),
        new ParallelGroup(
            Intake.INSTANCE.up(),
            Intake.INSTANCE.spinStop()
        ),
        new Delay(0.1),
        Arm.INSTANCE.retract(),
        new Delay(0.1)
    );
}

public Command transfer() {
    return new SequentialGroup(
        Arm.INSTANCE.transfer1(),
        new Delay(0.15),
        Arm.INSTANCE.transfer2(),
        new Delay(0.15),
        Arm.INSTANCE.transfer3(),
        new Delay(0.3),
        Arm.INSTANCE.transfer4(),
        new Delay(0.35),
        new ParallelGroup(
            Arm.INSTANCE.transfer5(),
            Arm.INSTANCE.extend()
        ),
        new Delay(0.15),
        new ParallelGroup(
            Arm.INSTANCE.transfer6(),
            Lift.INSTANCE.toBucket()
        ),
        new Delay(0.1),
        Arm.INSTANCE.transfer7(),
        new Delay(0.1),
        Arm.INSTANCE.bucketScore(),
        new Delay(0.2),
        Arm.INSTANCE.goToIdle(),
        new Delay(0.15),
        Lift.INSTANCE.toBottom()
    );
}
```

2024-2025 INTO THE DEEP



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